Questions

Q1.

*Many factors can affect the stopping distance of a car.

Some of these factors involve the driver and some of these factors involve the car or the road.

Explain how the stopping distance of a car is affected by

- factors involving the driver
- factors involving the car or the road.You should include examples in your explanations.

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(Total for question = 6 marks)

Q2.

Shot-put is an Olympic event.
The shot is a heavy ball.
An athlete throws the shot as far as possible.

A sports scientist analyses an athlete's throw to help improve performance.

The scientist takes pictures of the athlete every 0.1 s during one throw.

Figure 13 shows the pictures of one throw.

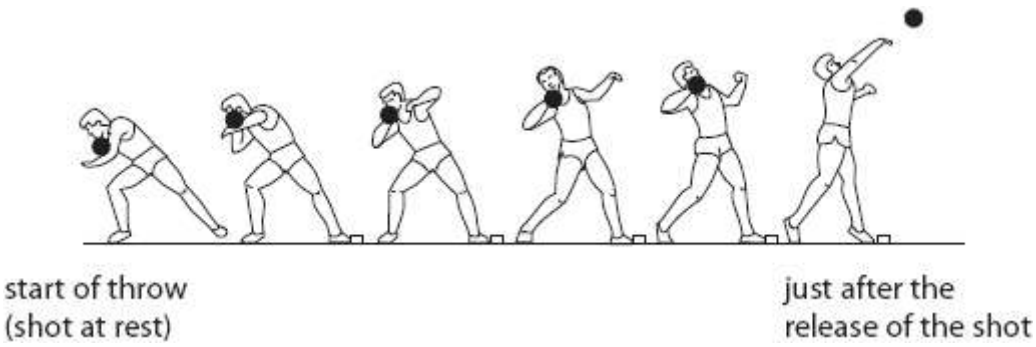

Figure 13

(i) Estimate the amount of time during the throw when the shot is in the athlete's hand.

(1)

time = ... s

(ii) Explain how the scientist could improve this method of analysing the throw.

(2)

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(iii) The average acceleration of the shot while in the athlete's hand is 20.6 m/s^2.

The mass of the shot is 7.26 kg.
Calculate the average force that the athlete applies to the shot during the throw.

(2)

force = ... N

(iv) In another throw, the shot is in the athlete's hand for 0.48 s.

The average acceleration during this time is 23 m/s^2.
Calculate the velocity of the shot as it leaves the athlete's hand.

(3)

velocity = ... m/s

(Total for question = 8 marks)

Q3.

Quantities can be either scalar or vector.

Figure 7 shows a ball bearing as it falls slowly through a clear, dense liquid.

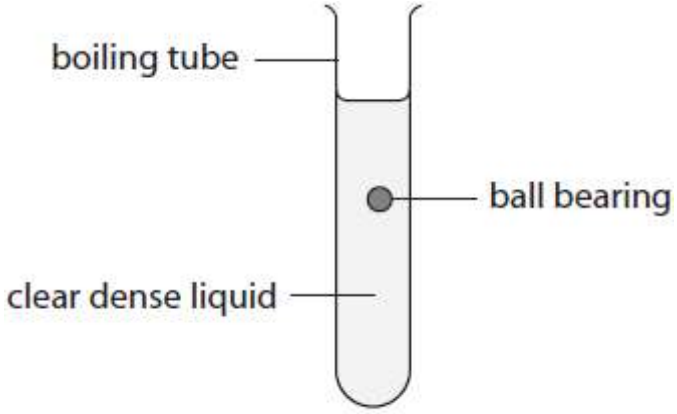

Figure 7

The apparatus in Figure 7 is used to find the average speed of the ball bearing as it falls.

(i) Devise an experiment to determine the average speed of the ball bearing as it falls through the liquid.

(4)

You should include:

- any extra apparatus you would use to take measurements
- the measurements you would take
- how you would calculate the speed.

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(ii) A student thinks that the ball bearing falls through the liquid at a constant speed.

Explain how you could develop this experiment to determine if the ball bearing falls through the liquid at constant speed.
You may draw a diagram to help your answer.

(2)

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(Total for question = 6 marks)

Q4.

Figure 7 shows a skier going down a hill.

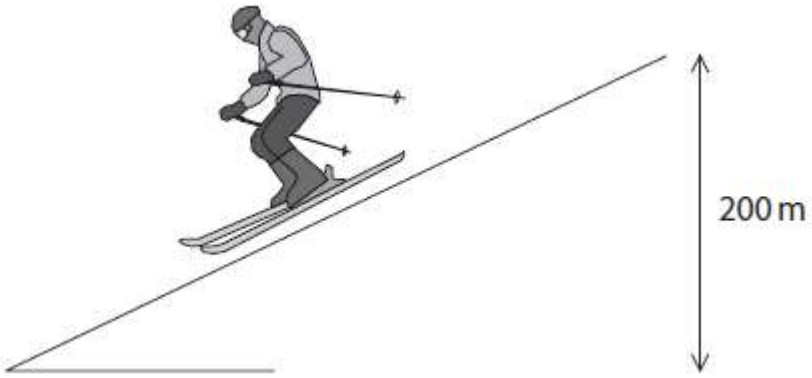

Figure 7

Describe how her speed at the bottom of the slope could be determined.

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(Total for question = 3 marks)

Q5.

Figure 16 is a velocity/time graph for a toy train on a straight track for 7 seconds.

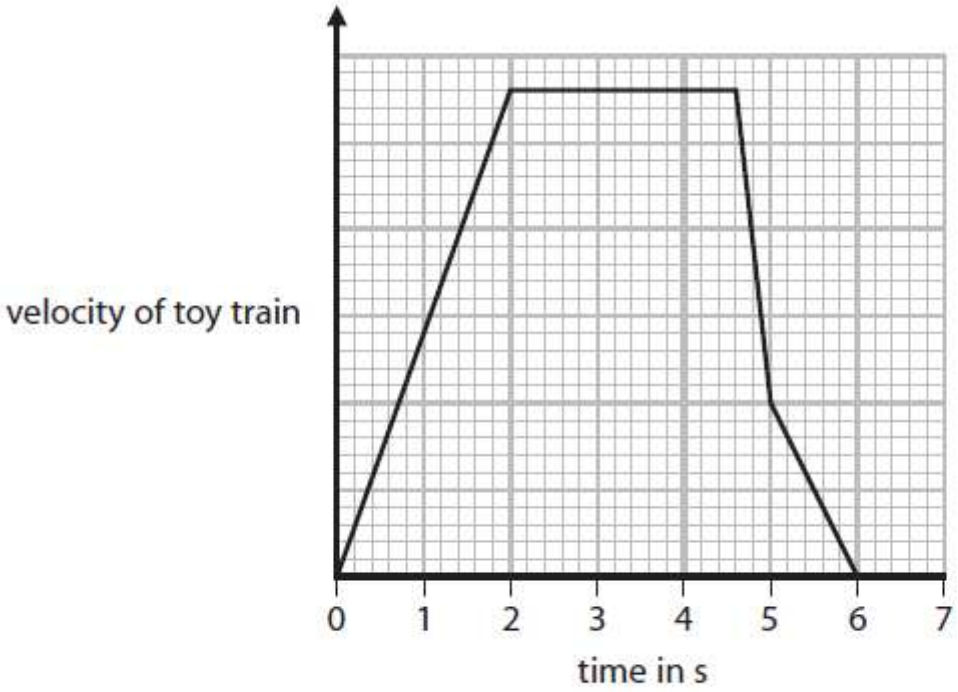

Figure 16

Using information from the graph, describe when and how the velocity and acceleration of the toy train change at different times during the 7 seconds shown on the graph.

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(Total for question = 6 marks)

Q6.

A student lifts a toy car from a bench and places the toy car at the top of a slope as shown in Figure 16.

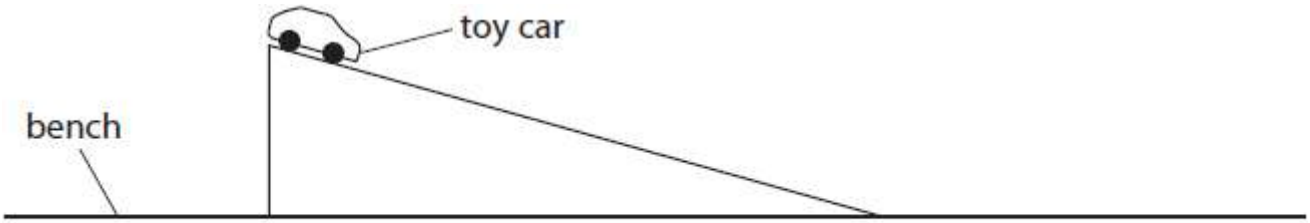

Figure 16

The student lets the toy car roll down the slope.

Describe how the student could find, by experiment, the speed of the toy car at the bottom of the slope.

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(Total for question = 4 marks)

Q7.

Some technicians are investigating factors affecting the braking distance of a bicycle.
They start by investigating how the speed of a bicycle affects the braking distance.

Figure 8 is a graph of their results.

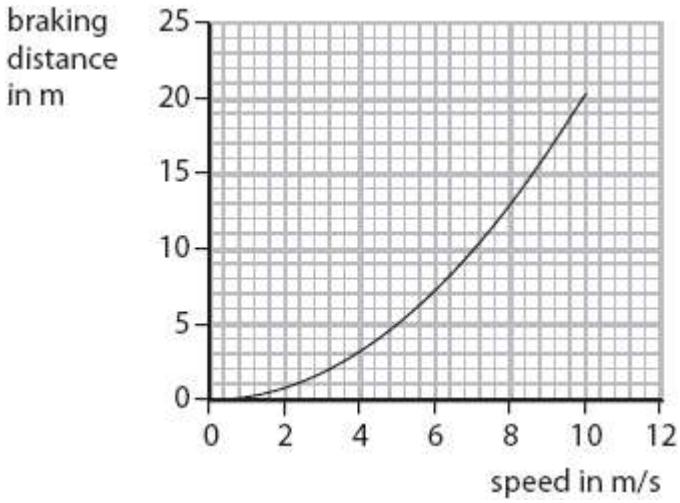

Figure 8

(i) Use the information in Figure 8 to describe how the braking distance of this bicycle changes with

speed.

(2)

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(ii) Figure 8 shows readings taken on a **dry** road.

Draw a curve on Figure 8 to show how the braking distance would change with speed for a bicycle on a **wet** road.

(2)

(iii) Describe how the technicians could develop their investigation to see how the mass of the bicycle would affect the braking distance.

(3)

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(Total for question = 7 marks)

Q8.

A student has a trolley and a ramp, as shown in Figure 21.

The height, H, of one end of the ramp can be adjusted.

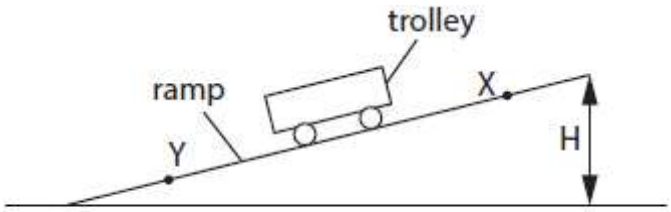

Figure 21

The student investigates how the average speed of the trolley between X and Y depends on the height, H, of the ramp.

Describe

- the additional equipment that the student needs
- how that equipment is used to obtain the measurements needed.

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(Total for question = 6 marks)

Q9.

Figure 5 shows the apparatus a student uses to investigate how the stopping distance of a toy car depends on the type of surface that it is stopping on.

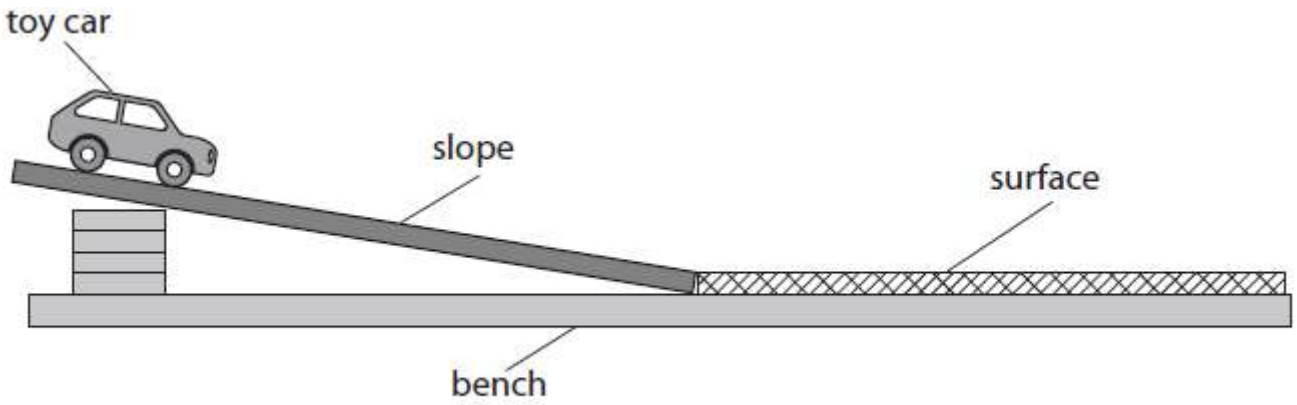

Figure 5

Describe an experiment to find out how the stopping distance depends on the surface that stops the toy car.

(2)

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(Total for question = 2 marks)

Q10.

(i) The car is moving at 90 km/h when the driver has to stop.

Calculate the thinking time of the driver.
Using the equation:
time = distance ÷ average speed

(2)

thinking time = ... s

(ii) A car has a mass of 1300 kg.

Calculate the kinetic energy of the car when it is travelling at 20 m/s.

(2)

kinetic energy = ... J

(Total for question = 4 marks)

Mark Scheme

Q1.

Question number	Indicative content	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 strand 1 (6 marks)** factors concerning driver • change in reaction time • tiredness • effect of drugs • type of footwear • how hard the driver presses the pedal effect of any of the above on stopping distance, • increased stopping distance • increased thinking distance • increased reaction time factors concerning car or road • mass / weight of car • speed of car • state of brakes • state of tyres • state of road effect of any of the above on stopping distance, e.g. • increased thinking/braking distance • increased stopping distance	**(6)**

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

SUMMARY, for guidance			
Level	**Mark**	**Additional Guidance**	**General additional guidance – the decision within levels** e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Elements of physics, i.e. isolated factor(s) about **either** car **or** driver	Possible candidate responses worn tyres / tired driver worn tyres and icy road
Level 2	3–4	Additional guidance Some understanding shown, i.e. **either** link between factor and effect **or** a driver factor **and** a car factor	Possible candidate responses worn tyres cause increased stopping distance. **or** worn tyres and tired driver
Level 3	5–6	Additional guidance Understanding is detailed and fully developed, i.e. link between factor and effect - both for driver **AND** for car	Possible candidate responses worn tyre causes increased stopping distance. **and** tired driver causes increased stopping distance

Q2.

Question number	Answer	Additional guidance	Mark
(i)	0.45 (s) (1)	Allow any value ≥ 0.4 and ≤ 0.5	**(1)**

Question number	Answer	Additional guidance	Mark
(ii)	An explanation that combines improvement of the experimental procedure (1 mark) and justification/reasoning which must be linked to the improvement (1 mark) • take pictures more frequently (1) • in order to determine exact time of the release. (1)	other responses may be acceptable	(2)

Question number	Answer	Additional guidance	Mark
(iii)	Substitution (1) F = 7.26 x 20.6 Evaluation (1) 150 (N)	Accept 149.6 (N) full marks will be awarded for correct numerical answer without working	(2)

Question number	Answer	Additional guidance	Mark
(iv)	Rearrangement (1) v = a x t Substitution (1) v = 23 x 0.48 Evaluation (1) 11 (m/s)	Accept 11.04(m/s) full marks will be awarded for correct numerical answer without working	(3)

Q3.

Question number	Answer	Additional guidance	Mark
(i)	A plan including four of the following measurement of appropriate distance (1) measurement of appropriate time (1) use of speed = $\frac{\text{distance}}{\text{Time}}$ (1) detail (1) e.g. repeat and average, use ruler/stop clock, mark a line near the top and bottom of liquid		**(4)** **AO3**

Question number	Answer	Additional guidance	Mark
(ii)	An explanation linking **two** from: add more lines (at equal distances)(1) measure the time of fall for each distance (1) compare the times (1)	 use longer test tube / use different heights of liquid / use different sections of the liquid e.g. {equal times =constant speed} / {shorter time = acceleration}	**(2)** **AO3**

Q4.

Question Number	Answer	Additional guidance	Mark
	a description to include 3 points from: • measure a distance (at the bottom) / use mark(er)s (certain distance apart) (1) • starting timer (at first mark(er)) (1) • stopping timer (at 2nd mark(er)) OR measures a time (interval) (1) • (use speed) = distance/time (1)	use a light gate (or equivalent sensors idea) not over whole slope for this mark point use of video / (speed) camera /interrupts the light beam accept any time measured for this mp including data logger OR timer / stopwatch	(3) AO 2 2

Q5.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO3** • graph starts at zero • graph increases to a maximum at 2 s • graph stays constant for 2.6 s • graph decreases to zero at 6 s • graph stays at zero after 6 s • graph decreases steeply until 5 s • graph decreases less steeply until 6 s • graph at zero between 6 and 7s **AO2** • velocity is zero at time zero • velocity increases/train accelerates until 2 s • velocity is constant for 2.6 s • velocity decreases/train decelerates until 6 s • deceleration changes at 5 s • acceleration is gradient of graph • velocity zero between 6 and 7 s	**(6)** **AO2** **AO3**

Level	Mark	Descriptor
	0	• No awardable content
Level 1	1–2	• Interpretation and evaluation of the information attempted but will be limited with a focus on mainly just one variable. Demonstrates limited synthesis of understanding. (AO3) • The description attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3–4	• Interpretation and evaluation of the information on both variables, synthesising mostly relevant understanding. (AO3) • The description is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5–6	• Interpretation and evaluation of the information, demonstrating throughout the skills of synthesising relevant understanding. (AO3)
		• The description is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated facts about the movement of the train or the shape of the graph	Possible candidate responses the train speeds up and slows down
Level 2	3–4	Additional guidance Description of motion in at least 2 parts of the graph. At least one of those parts linked to data from the graph.	Possible candidate responses the train speeds up for the first 2 seconds then stays at a constant speed
Level 3	5–6	Additional guidance Description of motion in at least 3 parts of the graph. At least two of those parts linked to data from the graph.	Possible candidate responses the train speeds up for the first 2 seconds then stays at a constant speed for 2.6 seconds then slows down

Q6.

Question number	Answer	Additional guidance	Mark
	A description to include:		**(4)**
	measurement of (relevant) distance (1)	one of distance down slope or distance along bench or length of toy car/card	
	measurement of (relevant) time (1)	'record the distance the car travels and time it' scores 2 marks	
	use of speed = $\frac{\text{distance}}{\text{time}}$ (1)		
	detail (1)	for example: speed down slope × 2 mark distance along bench use a light gate speed gun at the bottom of the slope repeating AND averaging	

Q7.

Question number	Answer	Additional guidance	Marks
(i)	An answer that combines points of interpretation/evaluation to provide a logical description: • braking distance increases as speed increases (1) • but increases at an increasing rate (1)	 accept takes 'longer' to brake accept positive correlation allow any clear correct idea of non-linearity / **use** of data from the graph indicating this trend	**(2)**

Question number	Answer	Additional guidance	Marks
(ii)	• correctly shaped curve (from the origin) with an increasing gradient (1) • above / between existing curve and y-axis (1)	reject if the line wavers significantly (up to 10 m/s) independent mark ignore tramlining ignore anything after speed = 10 m/s	**(2)**

Question number	Answer	Additional guidance	Marks
(iii)	an answer that combines 3 of the following points to provide a logical description of **the method**:	**must** be in the context of investigating **ignore repetition of stem of question**	**(3)**
	• add / change masses (or weights) to the bike (1)	bikes of different masses	
	• **measure** the braking distance (1)	however 'measure' is expressed	
	• use the same (initial) speed (1) • use same braking force/cyclist (1)	ignore 'repeating the experiment', without changing masses / weights	

Q8.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive, and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **measurement of distance using a rule or similar to find** • height of the ramp • travelled length of ramp / distance x to Y • width of card (if used) • distance between light gates (if used) • distance between dots on tape (if used) **measurement of time such as** • use of ticker-tape • use of a single light gate connected to electronic timer with a card fixed to the trolley • use of two light gates connected to electronic timer with a means of interrupting the light beams • use of (manually operated) stop clock /watch / timer	**6** **AO1.2**

	determination of speed • detail about which distance and time measurements are being used • repeat and average • repeat using different heights of the ramp	

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) • Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) • Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) • Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Limited description of measurement of distance and time	Possible candidate responses Measure the distance down the ramp. determine the time it took.
Level 2	3–4	Additional guidance Partial description including **two** from measurement of a relevant distance measurement of relevant time calculation of speed	Possible candidate responses Use the light gates to measure the time it took to go down. Measure the distance down the ramp with a ruler.
Level 3	5–6	Additional guidance Detailed description of measurements of relevant distances and associated times to find speed	Possible candidate responses Use light gates to measure the time to go from X to Y Use a ruler to measure the distance between the light gates. Divide distance between the light gates by the time taken to travel from X to Y.

Q9.

Question number	Answer	Additional guidance	Mark
	A description including two from let the car roll down the slope from the same point on the slope (1) measure distance it travels (along horizontal surface) (1) change the surface/ use different surfaces (1)	 see how far it travels allow time it takes to stop	(2) AO1

Q10.

Question number	Answer	Additional guidance	Mark
(i)	substitution Time = 37/ 25 (1) Evaluation (1) = 1.5 (s)	 Allow 1.48 (s) full marks will be awarded for correct numerical answer without working	(2)

Question number	Answer	Additional guidance	Mark
(ii)	substitution K.E. = 0.5 x 1300 x 20^2 (1) evaluation (1) = 260,000 J	 260 kJ full marks will be awarded for correct numerical answer without working	(2)